

DRY-RUN GUIDE | selection.py
Source: Dsa\sort\selection.py

CORE BEHAVIOR

Function-only selection sort: repeatedly finds the smallest remaining value, swaps it into the next final position, and returns the sorted list; importing the module prints nothing.

REPRESENTATIVE INPUT

Call `selection_sort([4,1,3,2])`.

STEP TRACE

`i=0` finds min at index 1; swap -> `[1,4,3,2]`. `i=1` finds index 3; swap -> `[1,2,3,4]`. Later passes leave the suffix ordered.

EXPECTED OUTPUT / RETURN

Return value: `[1,2,3,4]`; no console output.

ACTUAL SOURCE EXCERPT

Verbatim excerpt; line numbers refer to the original source, which was not modified.

```
001 | def selection_sort(arr):  
002 |     n = len(arr)  
003 |     for i in range(n):  
004 |         min_idx = i  
005 |         for j in range(i + 1, n):  
006 |             if arr[j] < arr[min_idx]:  
007 |                 min_idx = j  
008 |             arr[i], arr[min_idx] = arr[min_idx], arr[i]  
009 |     return arr
```

DRY-RUN NOTE

Trace describes actual source behavior, including errors where applicable.